



Recommending Beauty Products on Amazon

Using Cross-Categorical Reviews and Multimodal Item Content Data

Project 2 Proposal

CS608: Recommender Systems

Group 3

Gabriel Manuel Sidik, Jolene Yeo Gin Ning, Ngin Yun Chuan, Wang Yufang, Yudhistira Onggowarsito

The Cold-Start Problem in Beauty Product Recommendations

Recommending *Beauty* products is a high-margin profit driver for Amazon.

However, traditional Collaborative Filtering (CF) models trained on Amazon product reviews struggle to provide personalized recommendations for many shoppers due to data sparsity.

72%

% of *Beauty* shoppers that have **<5 reviews** in that category

99.9998%

Sparsity of the user-item interaction matrix for *Beauty* reviews



Key Insight - Use Cross-Category & Multimodal Data

What if we learnt user preferences from *adjacent categories* and *product descriptions & images* to predict their next Beauty purchase *earlier*?

Beauty-adjacent categories contain user preferences:

CLOTHING

Aesthetic taste & style preference

SPORTS & OUTDOORS

Lifestyle orientation



4 Complementary Data Modalities

01

Cross-Category Interaction Data

Map user-item interactions across Beauty, Clothing, and Sports.

02

Visual Data from Product Images

Extract visual embeddings from images to predict style and taste.

03

Temporal Data from Product Reviews

Captures sequential, time-aware purchase patterns.

04

Text and Semantic Product Metadata

Enriches understanding via descriptions, features, and pricing.

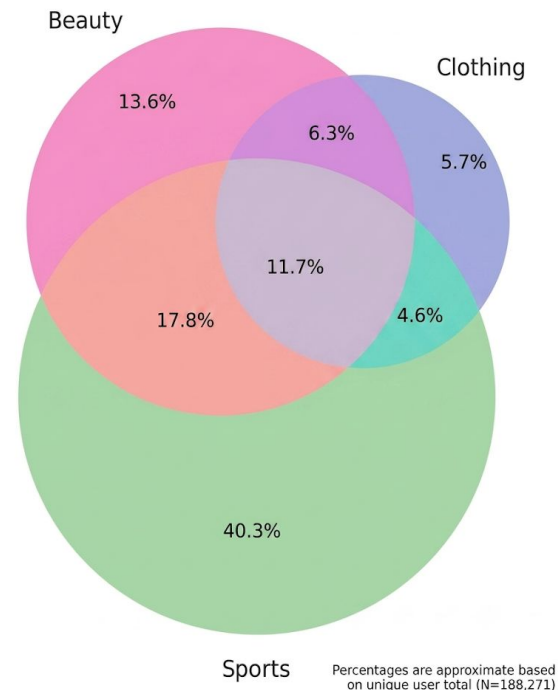
The Dataset

[McAuley-Lab/Amazon-Reviews-2023](#)¹

Total dataset: 571.54M user-product reviews on Amazon from 1996 to 2023,
grouped by category

	Users	Products	Reviews
 Beauty Target domain	11.3M	1.0M	23.9M
 Clothing Source domain	22.6M	7.2M	66.0M
 Sports & Outdoors Source domain	10.3M	1.6M	19.6M

User Overlap Across Categories (percentages)



¹ [2403.03952] Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders

Data Fields

User Review

- **user_id** - ID of the reviewer
- **parent_asin** - ID of the product
- **rating** - rating of the product from 1-5
- **title** - title of the review
- **text** - text body of the review
- **images** - images the user post after receiving the product
- **timestamp** - time of the review in unix time
- **verified_purchase** - whether the user purchase is verified (true/false)
- **helpful_vote** - no. of helpful votes of the review

Product

- **parent_asin** - ID of the product
- **title** - name of the product
- **description** - description of the product
- **features** - bullet point format features of the product
- **price** - price of the product in USD
- **images** - images of the product
- **store** - store name of the product
- **categories** - hierarchical categories of the product
- **details** - product details including materials, brand, size, etc
- **average_rating** - average rating on the product
- **rating_number** - no. of ratings on the product

The RecSys Problem

Problem Definition

We define this as a **sequential recommendation problem** for the *Beauty* category. Given a user's historical sequence of Amazon reviews in Beauty and Beauty-adjacent categories, we predict the next Beauty product the user will buy (review).

User Cohort Segmentation

Cold Users (~72%)

< 5 Beauty reviews

Warm Users (~28%)

>= 5 Beauty reviews

Evaluation Methodology

DATASET SPLIT

Temporal Segmentation

We segment the dataset temporally with two specific timestamp cut-offs for a 8:1:1 train-validation-test split¹.

These specific cut-offs are shared across all categories.

METRICS

NDCG@10 & Recall@10

Primary performance indicators used to measure ranking quality and retrieval effectiveness.

User Segmentation

COLD USERS
Beauty

WARM USERS
Beauty

Reported separately for cold and warm user cohorts.

¹ This split is similar to [2403.03952] Bridging Language and Items for Retrieval and Recommendation: Benchmarking LLMs as Semantic Encoders

Approaches: Encoding Multimodal Item Content

1

Product Metadata

e.g., price, store

1. Do z-score normalization, bucketization (numerical)
2. Treat as product text

2

Product Text

e.g., features, details

1. Encode with pretrained text encoders, e.g., BERT, RoBERTa
2. Encode with LLM embedding models, e.g., Qwen3, GritLM, Mistral, Gemini
3. Encode with VLMs, e.g., CLIP

3

Product Images

1. Encode with pretrained CNNs
e.g. ResNet
2. Convert to text with VLMs,
e.g., ViT-GPT2, BLIP-2,
Florence-2
3. Encode with VLMs, e.g., CLIP

Models & Experiments

1

Traditional CF Baseline

WMF, LightGCN, etc on Beauty reviews, product IDs only

2

Sequential CF Baseline

SASRec, GRU4Rec, BERT4REc, etc on Beauty reviews, product IDs only

3

Sequential with Multimodal Item Content

SASRecF, GRU4RecF, etc on Beauty reviews only, with multimodal embeddings...

a

From text/numerical product data extracted with pretrained text encoder models/LLMs like BERT, RoBERTa, GritLM, Qwen3, Mistral, CLIP...

b

From visual product data extracted with pretrained CNN encoders/VLMs like BLIP-2, ViT-GPT2, Florence-2, CLIP...

4

Cross-Domain Sequential with Multimodal Item Content

EMCDR, LLM-EMF¹, etc on Beauty + Beauty-adjacent reviews with multimodal embeddings

¹ [2506.17966] LLM-Enhanced Multimodal Fusion for Cross-Domain Sequential Recommendation

Thanks!

Recommending Beauty Products on Amazon

Using Cross-Categorical Reviews and Multimodal Item Content Data

Project 2 Proposal

CS608: Recommender Systems

Group 3



Appendix

Data Fields: Availability

Based on a subset of Beauty and Clothing categories (*All_Beauty* + *Amazon_Fashion*)...

	field	source_column	present	rows	non_null	non_null_pct	non_empty	non_empty_pct	examples
0	Item descriptions	description	True	921435	921435	100.00	68665	7.45	['The Precision Plunger Bars are designed to ...
1	Item features	features	True	921435	921435	100.00	464336	50.39	['Material: 304 Stainless Steel; Brass tip', ...
2	Item prices	price	True	921435	52648	5.71	52648	5.71	[6.99, 86.95, 79.5]
3	Item images	images	True	921435	921435	100.00	921434	100.00	[{'thumb': 'https://m.media-amazon.com/images...
4	Item videos	videos	True	921435	921435	100.00	63946	6.94	[{'title': '2PC rainbow cloud bath bombs,Floa...
5	Item categories	categories	True	921435	921435	100.00	0	0.00	[]
6	Item details	details	True	921435	921435	100.00	890486	96.64	[{'Package Dimensions': '7.1 x 5.5 x 3 inches;...
7	Item bought together	bought_together	True	921435	0	0.00	0	0.00	[]

Data Fields: Examples

	parent_asin	description
4	B07NGFDN6G	[The Precision Plunger Bars are designed to work seamlessly with the Precision Disposable 1. 25" Contoured Soft Cartridge Grips and the Precision Disposable 1" Textured Soft Cartridge Grips to drive cartridge needles with vice style or standard tattoo machine setups. These plunger bars are manufactured from 304 Stainless Steel and feature a brass tip. The plungers are sold in a bag of ten in your choice of 88mm, 93mm, or 98mm length.]
5	B07G9GWFSM	[Description, The false toenails are durable with perfect length. You have the option to wear them long or clip them short, easy to trim and file them to in any length and shape you like. Plus, ABS is kind of green enviromental material, and makes the nails durable, breathable, light even no pressure on your own toenails. Fit well to your natural toenails. Non toxic, no smell, no harm to your health., Feature, - Color: As Shown.- Material: ABS.- Size: 14.3 x 7.2 x 1cm., Package Including, 100 x Pieces fake toenails]
8	B01ERJEGS6	[Edt spray 3 oz design house: balmain]

	parent_asin	features
4	B07NGFDN6G	[Material: 304 Stainless Steel; Brass tip, Lengths Available: 88mm, 93mm, 98mm, Accepts cartridge needles with vice style tattoo machines, Works perfectly with Precision Disposable Soft Cartridge Grips, Price per one bag of 10 plungers]
5	B07G9GWFSM	[The false toenails are durable with perfect length. You have the option to wear them long or clip them short, easy to trim and file them to in any length and shape you like., ABS is kind of green enviromental material, and makes the nails durable, breathable, light even no pressure on your own nails., Fit well to your natural toenails. Non toxic, no smell, no harm to your health., Wonderful as gift for girlfriend, family and friends., The easiest and most efficient way to do your toenail tips for manicures or nail art designs. It's fashion, creative, a useful accessory brighten up your look, also as a gift.]
8	B01ERJEGS6	[Extatic Balmain Gold Musk By Balmain Edt Spray 3 Oz]

Data Fields: Examples

	parent_asin	images
0	B01CUPMQZE	[{'thumb': 'https://m.media-amazon.com/images/I/41qfjSfqNyL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/41qfjSfqNyL.jpg', 'variant': 'MAIN', 'hi_res': None}, {'thumb': 'https://m.media-amazon.com/images/I/41w2yznfuZL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/41w2yznfuZL.jpg', 'variant': 'PT01', 'hi_res': 'https://m.media-amazon.com/images/I/71i77Aul9xL._SL1500_.jpg'}]
1	B076WQZGPM	[{'thumb': 'https://m.media-amazon.com/images/I/41b+11d5igL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/41b+11d5igL.jpg', 'variant': 'MAIN', 'hi_res': 'https://m.media-amazon.com/images/I/71g1lP0pMbL._SL1500_.jpg'}, {'thumb': 'https://m.media-amazon.com/images/I/41j2ocUzCtL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/41j2ocUzCtL.jpg', 'variant': 'PT01', 'hi_res': 'https://m.media-amazon.com/images/I/81OqvR94isL._SL1500_.jpg'}]
2	B000B658RI	[{'thumb': 'https://m.media-amazon.com/images/I/31bz+uqzWCL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/31bz+uqzWCL.jpg', 'variant': 'MAIN', 'hi_res': None}, {'thumb': 'https://m.media-amazon.com/images/I/31bz+uqzWCL._SS40_.jpg', 'large': 'https://m.media-amazon.com/images/I/31bz+uqzWCL.jpg', 'variant': 'PT01', 'hi_res': None}]

	parent_asin	details
0	B01CUPMQZE	{'Package Dimensions': '7.1 x 5.5 x 3 inches; 2.38 Pounds', 'UPC': '617390882781'}
1	B076WQZGPM	{'Item Form': 'Powder', 'Skin Type': 'Acne Prone', 'Brand': 'Yes To', 'Age Range (Description)': 'Adult', 'Unit Count': '10 Fl Oz', 'Is Discontinued By Manufacturer': 'No', 'Item model number': 'SG_B076WQZGPM_US', 'UPC': '653801351125', 'Manufacturer': 'Yes to Tomatoes'}
2	B000B658RI	{'Manufacturer': 'Levine Health Products'}